

**第9次习题课** 2025年11月28日周五晚7:20 六教6B206(王嘉濠助教), 6B207(苑存远助教). 学习闭形式和Green公式的应用, 学习第二型曲线积分的计算.

**【习题1】** 设  $n \geq 2, \Omega \subset \mathbb{R}^n$  为开集,  $F = (F^1, F^2, \dots, F^n)^\tau \in C^1(\Omega, \mathbb{R}^n)$  是一个无旋场, 即  $F'(x) = F'(x)^\tau \quad \forall x \in \Omega$ , 即

$$\frac{\partial F^j(x)}{\partial x^i} = \frac{\partial F^i(x)}{\partial x^j} \quad \forall i, j \in \{1, 2, \dots, n\}, \quad \forall x \in \Omega.$$

也即1-形式

$$\omega(x) = \sum_{i=1}^n F^i(x) dx^i, \quad x \in \Omega,$$

是  $\Omega$  上的一个闭形式.

设  $H \in C^2([0, 1]^2, \Omega)$  是一个  $C^2$  类的定端同伦, 即满足  $H(1, t) = H(s, 0) = H(s, 1) = x_0$  for all  $s, t \in [0, 1]$  其中  $x_0 \in \Omega$  为一个定点. 则对于闭合道路  $\gamma(t) := H(0, t), t \in [0, 1]$ , 有

$$\int_0^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt = 0.$$

**【证】** [证法1 (by 2020级张逸尘)]: 写

$$H(s, t) = (H^1(s, t), H^2(s, t), \dots, H^n(s, t)).$$

则拉回  $H^*\omega$  是  $[0, 1]^2 \subset \mathbb{R}^2$  上的一个  $C^1$  类的1次形式(见讲义第十一章例11.6.4):

$$H^*\omega(s, t) = \left( \sum_{i=1}^n F^i(H(s, t)) \frac{\partial H^i(s, t)}{\partial s} \right) ds + \left( \sum_{i=1}^n F^i(H(s, t)) \frac{\partial H^i(s, t)}{\partial t} \right) dt.$$

由此和假设可见,  $H^*\omega$  在边界  $\partial[0, 1]^2$  上的四段(顺时针排列)  $\{0\} \times [0, 1], [0, 1] \times \{1\}, \{1\} \times [0, 1], [0, 1] \times \{0\}$  上的表达式分别为

$$(H^*\omega)(0, t) = \sum_{i=1}^n F^i(H(0, t)) \frac{\partial H^i(0, t)}{\partial t} dt = \langle F(\gamma(t)), \gamma'(t) \rangle dt, \quad t \in [0, 1],$$

$$H^*\omega(s, 1) = \sum_{i=1}^n F^i(H(s, 1)) \frac{\partial H^i(s, 1)}{\partial s} ds = 0, \quad s \in [0, 1],$$

$$(H^*\omega)(1, t) = \sum_{i=1}^n F^i(H(1, t)) \frac{\partial H^i(1, t)}{\partial t} dt = 0, \quad t \in [0, 1],$$

$$H^*\omega(s, 0) = \sum_{i=1}^n F^i(H(s, 0)) \frac{\partial H^i(s, 0)}{\partial s} ds = 0, \quad s \in [0, 1].$$

因此有

$$\begin{aligned} \int_0^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt &= \int_0^1 (H^*\omega)(0, t) = \int_{\{0\} \times [0, 1]} H^*\omega \\ &= \int_{\{0\} \times [0, 1]} H^*\omega + \int_{[0, 1] \times \{1\}} H^*\omega + \int_{\{1\} \times [0, 1]} H^*\omega + \int_{[0, 1] \times \{0\}} H^*\omega \\ &= \int_{\partial([0, 1]^2)} H^*\omega \end{aligned}$$

(上面是按 $\partial[0, 1]^2$ 的顺时针方向积分)

$$\begin{aligned} &= - \int_{-\partial[0, 1]^2} H^* \omega \quad (\text{注意 } -\partial[0, 1]^2 \text{ 的方向是逆时针方向}) \\ &= - \int_{[0, 1]^2} dH^* \omega = - \int_{[0, 1]^2} H^* d\omega = - \int_{[0, 1]^2} 0 = 0. \end{aligned}$$

这里用到 $[0, 1]^2$ 上的Green公式, 拉回的微分运算 $dH^* \omega = H^* d\omega$ ,  $d\omega = 0$  和  $H^* 0 = 0$ .

[证法2 (by 2020级陈彦谔)]: 来证明函数

$$s \mapsto I(s) := \int_0^1 \langle F(H(s, t)), \frac{\partial}{\partial t} H(s, t) \rangle dt, \quad s \in [0, 1]$$

为常数, 由此可导出上述积分为零. 为此我们使用积分号下求导定理(不难验证这里满足该定理的条件) 得到

$$I'(s) = \int_0^1 \frac{\partial}{\partial s} \langle F(H(s, t)), \frac{\partial}{\partial t} H(s, t) \rangle dt, \quad s \in [0, 1].$$

对被积函数, 用对称性 $F'(x)^\tau = F'(x)$  我们计算

$$\begin{aligned} &\frac{\partial}{\partial s} \langle F(H(s, t)), \frac{\partial}{\partial t} H(s, t) \rangle \\ &= \langle \frac{\partial}{\partial s} (F(H(s, t))), \frac{\partial}{\partial t} H(s, t) \rangle + \langle F(H(s, t)), \frac{\partial^2}{\partial s \partial t} H(s, t) \rangle \\ &= \langle F'(H(s, t)) \frac{\partial}{\partial s} H(s, t), \frac{\partial}{\partial t} H(s, t) \rangle + \langle F(H(s, t)), \frac{\partial^2}{\partial s \partial t} H(s, t) \rangle \\ &= \langle [F'(H(s, t))]^\tau \frac{\partial}{\partial s} H(s, t), \frac{\partial}{\partial t} H(s, t) \rangle + \langle F(H(s, t)), \frac{\partial^2}{\partial s \partial t} H(s, t) \rangle \\ &= \langle \frac{\partial}{\partial s} H(s, t), F'(H(s, t)) \frac{\partial}{\partial t} H(s, t) \rangle + \langle F(H(s, t)), \frac{\partial^2}{\partial s \partial t} H(s, t) \rangle. \end{aligned}$$

而同样计算也有

$$\begin{aligned} &\frac{\partial}{\partial t} \langle F(H(s, t)), \frac{\partial}{\partial s} H(s, t) \rangle \\ &= \langle \frac{\partial}{\partial t} (F(H(s, t))), \frac{\partial}{\partial s} H(s, t) \rangle + \langle F(H(s, t)), \frac{\partial^2}{\partial t \partial s} H(s, t) \rangle \\ &= \langle F'(H(s, t)) \frac{\partial}{\partial t} H(s, t), \frac{\partial}{\partial s} H(s, t) \rangle + \langle F(H(s, t)), \frac{\partial^2}{\partial t \partial s} H(s, t) \rangle \\ &= \langle \frac{\partial}{\partial s} H(s, t), F'(H(s, t)) \frac{\partial}{\partial t} H(s, t) \rangle + \langle F(H(s, t)), \frac{\partial^2}{\partial t \partial s} H(s, t) \rangle. \end{aligned}$$

因

$$H \in C^2 \implies \frac{\partial^2}{\partial t \partial s} H(s, t) = \frac{\partial^2}{\partial s \partial t} H(s, t)$$

故得到

$$\frac{\partial}{\partial s} \langle F(H(s, t)), \frac{\partial}{\partial t} H(s, t) \rangle = \frac{\partial}{\partial t} \langle F(H(s, t)), \frac{\partial}{\partial s} H(s, t) \rangle, \quad (s, t) \in [0, 1]^2.$$

所以

$$I'(s) = \int_0^1 \frac{\partial}{\partial t} \langle F(H(s, t)), \frac{\partial}{\partial s} H(s, t) \rangle dt = \langle F(H(s, t)), \frac{\partial}{\partial s} H(s, t) \rangle \Big|_{t=0}^{t=1}$$

$$\begin{aligned}
&= \langle F(H(s, 1)), \frac{\partial}{\partial s} H(s, 1) \rangle - \langle F(H(s, 0)), \frac{\partial}{\partial s} H(s, 0) \rangle \\
&= \langle F(x_0), \mathbf{0} \rangle - \langle F(x_0), \mathbf{0} \rangle = 0, \quad s \in [0, 1].
\end{aligned}$$

所以  $I(s) \equiv \text{常数}$ . 最后再注意到  $H(1, t) \equiv x_0 \implies \frac{\partial}{\partial t} H(1, t) \equiv \mathbf{0} (t \in [0, 1])$  即得

$$\int_0^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt = I(0) = I(1) = \int_0^1 \langle F(H(1, t)), \frac{\partial}{\partial t} H(1, t) \rangle dt = 0.$$

□

**【注】** 证法2更基本, 因为它只用到积分号下求导定理和复合函数微分的链锁法则, 不涉及微分形式和Green公式. 这两种证明都要感谢单连通的**定量描述**, 即定端同伦. 没有这种量化描述就无法进行严格证明, 例如你无法仅靠文字描述“在 $\Omega$ 内连续收缩成一点”来严格证明相关性质.

**【习题2(习题1的应用)】** 设  $n \geq 2, \Omega \subset \mathbb{R}^n$  为一个单连通的开集. 则 $\Omega$ 上的光滑的1次闭形式也是恰当形式. 即若

$$\omega(x) = \sum_{i=1}^n F^i(x) dx^i, \quad x \in \Omega,$$

是 $\Omega$ 上的一个闭形式, 其中  $F^i \in C^1(\Omega), i = 1, 2, \dots, n$ , 则存在  $U \in C^2(\Omega)$  使得  $dU = \omega$  于 $\Omega$ , 即

$$\frac{\partial}{\partial x^i} U(x) = F^i(x), \quad x \in \Omega, i = 1, 2, \dots, n.$$

**【证】** 由 $\omega$ 是闭形式知  $F = (F^1, F^2, \dots, F^n)^T \in C^1(\Omega, \mathbb{R}^n)$  是一个无旋场, 即  $F'(x) = F'(x)^T \quad \forall x \in \Omega$ , 即  $\frac{\partial F^j(x)}{\partial x^i} = \frac{\partial F^i(x)}{\partial x^j} \quad \forall i, j \in \{1, 2, \dots, n\}, \quad \forall x \in \Omega$ .

取定一点  $x_0 \in \Omega$ , 定义函数  $U : \Omega \rightarrow \mathbb{R}$  如下:

$$U(x) = \int_0^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt, \quad x \in \Omega \quad (*)$$

其中  $\gamma \in \mathcal{P}_{x_0, x}$  这里

$$\begin{aligned}
\mathcal{P}_{x_0, x} &= \{ \gamma \in C([0, 1], \Omega) \mid \gamma(0) = x_0, \gamma(1) = x \text{ 且 } \gamma \text{ 在 } [0, 1] \text{ 上是分段 } C^1 \text{ 类的函数, 即存在} \\
&N \in \mathbb{N} \text{ 和 } 0 = t_0 < t_1 < \dots < t_{N-1} < t_N = 1 \text{ 使得 } \gamma|_{[t_{i-1}, t_i]} \in C^1([t_{i-1}, t_i], \mathbb{R}^n), i = 1, 2, \dots, N. \}
\end{aligned}$$

须证明  $U(x)$  是良好定义的, 即  $U(x)$  不依赖于道路  $\gamma \in \mathcal{P}_{x_0, x}$  的选择. 设  $\gamma_1, \gamma_2 \in \mathcal{P}_{x_0, x}$ . 令

$$\gamma(t) = \begin{cases} \gamma_1(2t) & \text{if } t \in [0, 1/2], \\ \gamma_2(2(1-t)) & \text{if } t \in [1/2, 1]. \end{cases}$$

则  $\gamma \in \mathcal{P}_{x_0, x_0}$  即  $\gamma$  是 $\Omega$ 内的一条闭路:  $\gamma(0) = \gamma(1) = x_0$ . 因 $\Omega$ 是单连通的, 故闭路 $\gamma$ 可以在 $\Omega$ 中连续缩成一点, 即存在连续映射  $H : [0, 1] \times [0, 1] \rightarrow \Omega$  满足定端同伦性质:

$$H(0, t) = \gamma(t), \quad H(1, t) = H(s, 0) = H(s, 1) = \gamma(0) \quad \forall s, t \in [0, 1].$$

我们将在第十三章证明一个多项式逼近性质, 这个逼近性质使我们可以假定 $H$ (从而 $\gamma$ )是充分光滑的, 例如可以假定 $H \in C^2([0, 1]^2, \Omega)$ . 于是据习题1有

$$\begin{aligned} 0 &= \int_0^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt = \int_0^{1/2} \langle F(\gamma(t)), \gamma'(t) \rangle dt + \int_{1/2}^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt \\ &= \int_0^{1/2} \langle F(\gamma_1(2t)), 2\gamma'_1(2t) \rangle dt + \int_{1/2}^1 \langle F(\gamma_2(2(1-t))), -2\gamma'_2(2(1-t)) \rangle dt \\ &= \int_0^1 \langle F(\gamma_1(t)), \gamma'_1(t) \rangle dt - \int_0^1 \langle F(\gamma_2(t)), \gamma'_2(t) \rangle dt \end{aligned}$$

所以

$$\int_0^1 \langle F(\gamma_1(t)), \gamma'_1(t) \rangle dt = \int_0^1 \langle F(\gamma_2(t)), \gamma'_2(t) \rangle dt.$$

所以 $U(x)$ 是良好定义的.

下证 $U \in C^2(\Omega)$ 且

$$\frac{\partial}{\partial x^i} U(x) = F^i(x), \quad x \in \Omega, i = 1, 2, \dots, n.$$

任取 $x \in \Omega$ . 因 $\Omega$ 是开集, 故存在 $\delta > 0$  使得 $B^n(x, \delta) \subset \Omega$ . 对任意 $i \in \{1, 2, \dots, n\}$ , 对任意 $0 < |h| < \delta$ , 取 $\gamma_0 \in \mathcal{P}_{x_0, x}$ , 令

$$\gamma(t) = \begin{cases} \gamma_0(2t) & \text{if } t \in [0, 1/2], \\ x + (2t - 1)h\mathbf{e}_i & \text{if } t \in [1/2, 1]. \end{cases}$$

则 $\gamma \in \mathcal{P}_{x_0, x+h\mathbf{e}_i}$ . 计算

$$\begin{aligned} U(x + h\mathbf{e}_i) &= \int_0^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt = \int_0^{1/2} \langle F(\gamma(t)), \gamma'(t) \rangle dt + \int_{1/2}^1 \langle F(\gamma(t)), \gamma'(t) \rangle dt \\ &= \int_0^{1/2} \langle F(\gamma_0(2t)), 2\gamma'_0(2t) \rangle dt + \int_{1/2}^1 \langle F(x + (2t - 1)h\mathbf{e}_i), 2h\mathbf{e}_i \rangle dt \\ &= \int_0^1 \langle F(\gamma_0(t)), \gamma'_0(t) \rangle dt + h \int_0^1 \langle F(x + th\mathbf{e}_i), \mathbf{e}_i \rangle dt \\ &= U(x) + h \int_0^1 F^i(x + th\mathbf{e}_i) dt \\ &\implies \frac{U(x + h\mathbf{e}_i) - U(x)}{h} = \int_0^1 F^i(x + th\mathbf{e}_i) dt \rightarrow F^i(x) \quad (h \rightarrow 0). \end{aligned}$$

所以 $\frac{\partial}{\partial x^i} U(x)$ 存在且 $\frac{\partial}{\partial x^i} U(x) = F^i(x)$ . 所以 $\frac{\partial}{\partial x^i} U = F^i$  于 $\Omega$ ,  $i = 1, 2, \dots, n$ . 因 $F^i \in C^1(\Omega)$ ,  $i = 1, 2, \dots, n$ , 故 $U \in C^2(\Omega)$ . 这就证明了 $dU = \omega$  于 $\Omega$ , 即 $U$ 是1次形式 $\omega$ 的一个原函数, 所以1次闭形式 $\omega$ 在单连通区域 $\Omega$ 上是恰当形式.  $\square$

### 【习题3】

(1) 设质点在力场 $F(x, y) = (a + \frac{e^x}{1+y^2})\mathbf{i} + \frac{2y(1-e^x)}{(1+y^2)^2}\mathbf{j}$  的作用下, 沿曲线 $x^2 + (y-1)^2 = 1$ 从 $(0, 0)$ 出发沿逆时针方向运动到点 $(1, 1)$ . 求在此过程中力场 $F$ 对质点所做的功 $W$ . 这里 $\mathbf{i} = (1, 0)$ ,  $\mathbf{j} = (0, 1)$ ;  $a$ 是常数.

(2) 已知力场  $F(x, y, z) = y\mathbf{i} - x\mathbf{j} + (x + y + z)\mathbf{k}$ , 求质点沿曲线  $\gamma(t) = (a \cos t, a \sin t, \frac{b}{2\pi}t)$  从点  $A = (a, 0, 0)$  运动到点  $B = (a, 0, b)$  时, 力场  $F$  对质点所做的功  $W$ . 这里  $\mathbf{i} = (1, 0, 0)$ ,  $\mathbf{j} = (0, 1, 0)$ ,  $\mathbf{k} = (0, 0, 1)$ ;  $a, b$  是常数.

【解】(1): 我们有

$$F(x, y) = (F^1(x, y), F^2(x, y)) = \left( a + \frac{e^x}{1 + y^2}, \frac{2(1 - e^x)y}{(1 + y^2)^2} \right).$$

设  $\Gamma$  为题中质点运行轨迹. 则所求之功为

$$W = \int_{\Gamma} F^1(x, y)dx + F^2(x, y)dy.$$

注意  $F^1, F^2 \in C^\infty(\mathbb{R}^2)$ . 计算

$$\frac{\partial F^2}{\partial x} = \frac{\partial}{\partial x} \left( \frac{2(1 - e^x)y}{(1 + y^2)^2} \right) = \frac{-2e^xy}{(1 + y^2)^2} = \frac{\partial F^1}{\partial y} \quad \forall (x, y) \in \mathbb{R}^2.$$

这表明 Jacobi 矩阵  $F'(x, y)$  是对称的 (此时  $F$  是列向量), 即  $F'(x, y) = F'(x, y)^T$ . 因此一次形式  $\omega := F^1dx + F^2dy$  是  $\mathbb{R}^2$  上的光滑的闭形式. 因  $\mathbb{R}^2$  是单连通的, 故由习题 2 及其证明知  $\omega$  有原函数  $U$ , 因此当  $(0, 0)$  确定后, 积分  $W = U(1, 1)$  就只由终点  $(1, 1)$  确定, 而与具有相同起点和终点的道路的选择无关. 为便于计算, 我们取汽车路线:  $\Gamma_* = \Gamma_1 \cup \Gamma_2$ , 其中  $\Gamma_1: x \in [0, 1], y = 0$ ;  $\Gamma_2: x = 1, y \in [0, 1]$ . 然后由积分的可加性有

$$\begin{aligned} W &= U(1, 1) = \int_{\Gamma_*} F^1(x, y)dx + F^2(x, y)dy \\ &= \int_{\Gamma_1} F^1(x, y)dx + F^2(x, y)dy + \int_{\Gamma_2} F^1(x, y)dx + F^2(x, y)dy \\ &= \int_0^1 F^1(x, 0)dx + \int_0^1 F^2(1, y)dy = \int_0^1 \left( a + \frac{e^x}{2} \right) dx + \int_0^1 \frac{2(1 - e)y}{(1 + y^2)^2} dy \\ &= a + \left( \frac{e^x}{2} \Big|_{x=0}^{x=1} \right) + (e - 1)(1 + y^2)^{-1} \Big|_{y=0}^{y=1} = a + \frac{e - 1}{2} + (e - 1) \frac{-1}{2} = a. \end{aligned}$$

(2): 用列向量表示我们有

$$F(x, y, z) = \begin{pmatrix} F^1(x, y, z) \\ F^2(x, y, z) \\ F^3(x, y, z) \end{pmatrix} = \begin{pmatrix} y \\ -x \\ x + y + z \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$\Rightarrow F'(x, y, z) =$  系数矩阵. 因这矩阵不是对称的, 即  $F'(x, y, z) \neq F'(x, y, z)^T$ , 所以  $F$  不是无旋场, 即一次形式  $\omega = F^1dx + F^2dy + F^3dz$  不是闭形式. 所以对功  $W$  的计算只能按第二型曲线积分的一般计算公式计算: 由题意知质点运行轨迹为  $\Gamma: \gamma(t) = (a \cos t, a \sin t, \frac{b}{2\pi}t), t \in [0, 2\pi]$ . 计算 (注意  $\overrightarrow{d(x, y, z)} := (dx, dy, dz)$  是切向量)

$$\begin{aligned} W &= \int_{\Gamma} \omega = \int_{\Gamma} \langle F(x, y, z), \overrightarrow{d(x, y, z)} \rangle \quad ((x, y, z) = \gamma(t)) \\ &= \int_0^{2\pi} \langle F(\gamma(t)), \gamma'(t) \rangle dt = \int_0^{2\pi} (F^1(\gamma(t))x'(t) + F^2(\gamma(t))y'(t) + F^3(\gamma(t))z'(t)) dt \end{aligned}$$

$$\begin{aligned}
&= \int_0^{2\pi} \left( a \sin t (-a \sin t) + (-a \cos t) a \cos t + \left( a \cos t + a \sin t + \frac{b}{2\pi} t \right) \frac{b}{2\pi} \right) dt \\
&= \int_0^{2\pi} \left( -a^2 + \left( \frac{b}{2\pi} \right)^2 t \right) dt = -a^2 2\pi + \frac{(2\pi)^2}{2} \left( \frac{b}{2\pi} \right)^2 = -2\pi a^2 + \frac{1}{2} b^2.
\end{aligned}$$

□

**【习题4】** 设实值或复值函数  $P, Q \in C^1(\mathbb{R}^2)$  满足

$$\int_{(x-x_0)^2+(y-y_0)^2=r^2, y \geq y_0} P(x, y) dx + Q(x, y) dy = 0 \quad \forall (x_0, y_0) \in \mathbb{R}^2, \forall r > 0.$$

证明  $P(x, y) \equiv 0, \frac{\partial}{\partial x} Q(x, y) \equiv 0$  于  $\mathbb{R}^2$ .

注: 复值函数的积分(被定义为)=该函数的实部积分加上i乘以虚部的积分. 对复值函数的实部和虚部分别使用Green公式可知Green公式对于复值函数也成立.

**【证】** 任取  $(x_0, y_0) \in \mathbb{R}^2, r > 0$ , 令

$$D_r = \{(x, y) \in \mathbb{R}^2 \mid (x - x_0)^2 + (y - y_0)^2 < r^2, y > y_0\},$$

$$\Gamma = \{(x, y) \in \mathbb{R}^2 \mid (x - x_0)^2 + (y - y_0)^2 = r^2, y \geq y_0\},$$

$$L = \{(x_0 + t, y_0) \mid t \in [-r, r]\}.$$

则沿闭路  $\Gamma \cup L = \partial D_r$  的逆时针方向积分, 由Green公式有

$$\begin{aligned}
&\int_{\overline{D_r}} \left( \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy = \int_{\partial D_r} P dx + Q dy \\
&= \int_{\Gamma} P dx + Q dy + \int_L P dx + Q dy = \int_L P dx + Q dy \\
&= \int_{-r}^r P(x_0 + t, y_0) dt = r \int_{-1}^1 P(x_0 + rt, y_0) dt
\end{aligned}$$

令

$$M = \max_{(x, y) \in \overline{D_1}} \left| \frac{\partial Q(x, y)}{\partial x} - \frac{\partial P(x, y)}{\partial y} \right|.$$

则  $M < \infty$  因此对任意  $0 < r < 1$  有

$$\frac{1}{2} \left| \int_{-1}^1 P(x_0 + rt, y_0) dt \right| \leq \frac{1}{2r} \int_{\overline{D_r}} \left| \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right| dx dy \leq M \frac{1}{2r} \frac{\pi r^2}{2} = M \frac{\pi r}{4} \rightarrow 0 \quad (r \rightarrow 0+)$$

所以(使用Lebesgue控制收敛定理或由  $P$  连续)

$$|P(x_0, y_0)| = \lim_{r \rightarrow 0+} \frac{1}{2} \left| \int_{-1}^1 P(x_0 + rt, y_0) dt \right| = 0.$$

由  $(x_0, y_0) \in \mathbb{R}^2$  的任意性知  $P(x, y) \equiv 0$  于  $\mathbb{R}^2$ .

将  $P \equiv 0$  代入上面等式得到

$$\int_{\overline{D_r}} \frac{\partial}{\partial x} Q(x, y) dx dy = 0.$$

据此得到

$$\begin{aligned} \left| \frac{\partial}{\partial x} Q(x_0, y_0) \right| &= \left| \frac{1}{m(\overline{D_r})} \int_{\overline{D_r}} \left( \frac{\partial}{\partial x} Q(x_0, y_0) - \frac{\partial}{\partial x} Q(x, y) \right) dx dy \right| \\ &\leq \frac{1}{m(\overline{D_r})} \int_{\overline{D_r}} \left| \frac{\partial}{\partial x} Q(x_0, y_0) - \frac{\partial}{\partial x} Q(x, y) \right| dx dy \\ &= \left| \frac{\partial}{\partial x} Q(x_0, y_0) - \frac{\partial}{\partial x} Q(x_r, y_r) \right| \rightarrow 0 \quad (r \rightarrow 0+) \end{aligned}$$

这里使用了积分中值定理, 其中  $(x_r, y_r) \in \overline{D_r}$ , 易见  $(x_r, y_r) \rightarrow (x_0, y_0) (r \rightarrow 0+)$ . 所以  $\frac{\partial}{\partial x} Q(x_0, y_0) = 0$ . 再由  $(x_0, y_0) \in \mathbb{R}^2$  的任意性知  $\frac{\partial}{\partial x} Q(x, y) \equiv 0$  于  $\mathbb{R}^2$ .  $\square$

【习题5】计算积分

$$I = \int_C \frac{e^y}{x^2 + y^2} \left( (x \sin x + y \cos x) dx + (y \sin x - x \cos x) dy \right)$$

其中  $C: x^2 + y^2 = 1$ , 取逆时针方向.

【解】令

$$P(x, y) = \frac{e^y}{x^2 + y^2} (x \sin x + y \cos x), \quad Q(x, y) = \frac{e^y}{x^2 + y^2} (y \sin x - x \cos x).$$

计算

$$\begin{aligned} \frac{\partial P}{\partial y} &= \frac{e^y(x^2 + y^2) - e^y 2y}{(x^2 + y^2)^2} (x \sin x + y \cos x) + \frac{e^y}{x^2 + y^2} \cos x \\ &= \frac{e^y}{x^2 + y^2} (x \sin x + y \cos x + \cos x) + \frac{-2e^y}{(x^2 + y^2)^2} (xy \sin x + y^2 \cos x), \\ \frac{\partial Q}{\partial x} &= \frac{e^y(-2x)}{(x^2 + y^2)^2} (y \sin x - x \cos x) + \frac{e^y}{x^2 + y^2} (y \cos x - \cos x + x \sin x) \\ &= \frac{-2e^y}{(x^2 + y^2)^2} (xy \sin x - x^2 \cos x) + \frac{e^y}{x^2 + y^2} (y \cos x - \cos x + x \sin x), \\ &\implies \\ \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} &= \frac{-2e^y}{(x^2 + y^2)^2} (y^2 + x^2) \cos x + \frac{e^y}{x^2 + y^2} 2 \cos x = 0. \end{aligned}$$

所以被积函数在  $\mathbb{R}^2 \setminus \{(0, 0)\}$  上是闭形式. 由Green公式得知对任意  $0 < \varepsilon < 1$ , 沿逆时针方向积分然后令  $\varepsilon \rightarrow 0^+$  得到

$$\begin{aligned} I &= \int_C \frac{e^y}{x^2 + y^2} \left( (x \sin x + y \cos x) dx + (y \sin x - x \cos x) dy \right) \\ &= \int_{x^2 + y^2 = \varepsilon^2} \frac{e^y}{x^2 + y^2} \left( (x \sin x + y \cos x) dx + (y \sin x - x \cos x) dy \right) \\ &\quad (x = \varepsilon \cos \theta, y = \varepsilon \sin \theta) \\ &= \frac{1}{\varepsilon^2} \int_0^{2\pi} e^{\varepsilon \sin \theta} \left[ (\varepsilon \cos \theta \sin(\varepsilon \cos \theta) + \varepsilon \sin \theta \cos(\varepsilon \cos \theta)) (-\varepsilon \sin \theta) + (\varepsilon \sin \theta \sin(\varepsilon \cos \theta) \right. \\ &\quad \left. - \varepsilon \cos \theta \cos(\varepsilon \cos \theta)) \varepsilon \cos \theta \right] d\theta \\ &= \int_0^{2\pi} e^{\varepsilon \sin \theta} \left( (\cos \theta \sin(\varepsilon \cos \theta) + \sin \theta \cos(\varepsilon \cos \theta)) (-\sin \theta) + (\sin \theta \sin(\varepsilon \cos \theta) \right. \end{aligned}$$

$$\begin{aligned}
& -\cos \theta \cos (\varepsilon \cos \theta) \cos \theta) \mathrm{d} \theta \\
& \rightarrow \int_0^{2 \pi}(\sin \theta(-\sin \theta)-\cos \theta \cos \theta) \mathrm{d} \theta=-2 \pi \quad\left(\varepsilon \rightarrow 0^{+}\right) .
\end{aligned}$$

所以  $I=-2 \pi$ .

□

**【习题6(不等号方向的确定)】** 计算下面的双纽线

$$\left(x^2+y^2\right)^2=a^2\left(x^2-y^2\right), \quad(x, y) \in \mathbb{R}^2$$

所围集合的面积, 其中  $a>0$ .

**【解】** 设双纽线包围的集合为  $E$ , 则

$$E=\left\{(x, y) \in \mathbb{R}^2 \mid\left(x^2+y^2\right)^2 \leq a^2\left(x^2-y^2\right)\right\} .$$

(思考一下为什么取不等号“ $\leq$ ”?). 令

$$E_{+}=\left\{(x, y) \in \mathbb{R}^2 \mid\left(x^2+y^2\right)^2 \leq a^2\left(x^2-y^2\right), x \geq 0\right\} .$$

则由对称性,  $E$  的面积等于

$$m(E)=2 m\left(E_{+}\right) .$$

取极坐标参数:  $x=r \cos \theta, y=r \sin \theta, \theta \in\left[-\pi / 2, \pi / 2\right]$ . 则有

$$(x, y) \in E_{+} \Longleftrightarrow(0 \leq) r^4 \leq a^2 r^2\left(\cos ^2 \theta-\sin ^2 \theta\right)=a^2 r^2 \cos (2 \theta) .$$

这给出

$$(x, y) \in E_{+} \Longleftrightarrow r \leq a \sqrt{\cos (2 \theta)}, \quad \theta \in\left[-\pi / 4, \pi / 4\right] .$$

所以由重积分的极坐标换元公式有

$$\begin{aligned}
m\left(E_{+}\right) & =\int_{-\pi / 4}^{\pi / 4} \int_0^{\infty} r \mathbf{1}_{\left\{r \leq a \sqrt{\cos (2 \theta)}\right\}} \mathrm{d} r \mathrm{d} \theta=\int_{-\pi / 4}^{\pi / 4} \int_0^{a \sqrt{\cos (2 \theta)}} r \mathrm{d} r \mathrm{d} \theta \\
& =\int_{-\pi / 4}^{\pi / 4} \frac{a^2 \cos (2 \theta)}{2} \mathrm{d} \theta=a^2 \int_0^{\pi / 4} \cos (2 \theta) \mathrm{d} \theta=a^2 \frac{\sin (2 \theta)}{2} \Big|_{\theta=0}^{\theta=\pi / 4}=\frac{a^2}{2} .
\end{aligned}$$

所以  $m(E)=2 m\left(E_{+}\right)=a^2$ .

**【另法】** 使用Green公式计算面积. 由Green公式有(沿逆时针方向积分)

$$m(E)=2 m\left(E_{+}\right)=\int_{\partial E_{+}} x \mathrm{d} y-y \mathrm{d} x .$$

对闭合曲线

$$\partial E_{+}=\left\{(x, y) \in \mathbb{R}^2 \mid\left(x^2+y^2\right)^2=a^2\left(x^2-y^2\right), x \geq 0\right\}$$

取极坐标参数:  $x=r \cos \theta, y=r \sin \theta, \theta \in\left[-\pi / 2, \pi / 2\right]$ . 则有

$$(x, y) \in \partial E_{+} \Longleftrightarrow(0 \leq) r^4=a^2 r^2\left(\cos ^2 \theta-\sin ^2 \theta\right)=a^2 r^2 \cos (2 \theta) .$$

这给出

$$(x, y) \in \partial E_+ \iff r = a\sqrt{\cos(2\theta)}, \quad \theta \in [-\pi/4, \pi/4].$$

所以 $\partial E_+$ 的极坐标参数表示为

$$x(\theta) = a\sqrt{\cos(2\theta)} \cos \theta, \quad y(\theta) = a\sqrt{\cos(2\theta)} \sin \theta, \quad \theta \in [-\pi/4, \pi/4].$$

注意 $x(\theta), y(\theta)$ 在 $\theta = \pm\pi/4$ 处不可导. 但在 $(-\pi/4, \pi/4)$ 内有

$$\begin{aligned} x(\theta)y'(\theta) - y(\theta)x'(\theta) &= (x(\theta))^2 \left( \frac{y(\theta)}{x(\theta)} \right)' = a^2 \cos(2\theta) \cos^2 \theta \left( \frac{\sin \theta}{\cos \theta} \right)' \\ &= a^2 \cos(2\theta) \cos^2 \theta \cdot \frac{1}{\cos^2 \theta} = a^2 \cos(2\theta). \end{aligned}$$

所以

$$\begin{aligned} \int_{\partial E_+} x dy - y dx &= \lim_{\varepsilon \rightarrow 0^+} \int_{-\pi/4+\varepsilon}^{\pi/4-\varepsilon} (x(\theta)y'(\theta) - y(\theta)x'(\theta)) d\theta = \lim_{\varepsilon \rightarrow 0^+} \int_{-\pi/4+\varepsilon}^{\pi/4-\varepsilon} a^2 \cos(2\theta) d\theta \\ &= \int_{-\pi/4}^{\pi/4} a^2 \cos(2\theta) d\theta = 2a^2 \int_0^{\pi/4} \cos(2\theta) d\theta = 2a^2 \frac{\sin(2\theta)}{2} \Big|_{\theta=0}^{\theta=\pi/4} = a^2. \end{aligned}$$

所以 $m(E) = a^2$ . □

**【习题7】** 设 $F \in C^2(\mathbb{R}^2)$  满足

$$D = \{(x, y) \in \mathbb{R}^2 \mid F(x, y) < 0\} \text{ 是有界单连通区域且 } \mathbf{0} = (0, 0) \in D,$$

$$|\nabla F(x, y)| \geq c \frac{(x^2 + y^2)^\alpha}{1 + x^2 + y^2} \quad \forall (x, y) \in \mathbb{R}^2$$

其中 $0 < \alpha < 1, c > 0$ 是常数. 令

$$\mathbf{n}(x, y) = \frac{\nabla F(x, y)}{|\nabla F(x, y)|}, \quad (x, y) \in \mathbb{R}^2 \setminus \{\mathbf{0}\}.$$

证明 $\partial D$ 的长度 $H_1(\partial D)$ 有限且可由下面公式计算:

$$H_1(\partial D) = \left| \int_D \nabla \cdot \mathbf{n}(x, y) dx dy \right|,$$

并且上式右端积分是绝对收敛的, 即

$$\int_D |\nabla \cdot \mathbf{n}(x, y)| dx dy < \infty.$$

这里 $\nabla \cdot = \text{div}$ 是散度算符.

**【证】** 由假知 $\mathbf{n} \in C^1(\mathbb{R}^2 \setminus \{\mathbf{0}\})$  且易见

$$\partial D = \{(x, y) \in \mathbb{R}^2 \mid F(x, y) = 0\}$$

并且在 $\partial D$ 上 $\mathbf{n}(x, y)$ 是 $\partial D$ 上的单位法向量.

设 $\varepsilon_0 > 0$ 充分小,  $0 < R < \infty$ 充分大使得 $\overline{B(\mathbf{0}, \varepsilon_0)} \subset D \subset B(\mathbf{0}, R)$ . 下设 $0 < \varepsilon < \varepsilon_0$ . 考虑有洞的区域:

$$D_\varepsilon = D \setminus \overline{B(\mathbf{0}, \varepsilon)}.$$

注意 $\frac{(x, y)}{\varepsilon}$ 是 $\partial B(\mathbf{0}, \varepsilon)$ 上的指向 $\overline{B(\mathbf{0}, \varepsilon)}$ 外部的单位法向量. 因此由有洞区域的Green公式的第一型积分表示知, 若 $\mathbf{n}(x, y)|_{\partial D}$ 在某一点从而处处指向 $\overline{D}$ 的外侧, 则

$$\begin{aligned} H_1(\partial D) &= \int_{\partial D} ds(x, y) = \int_{\partial D} \langle \mathbf{n}(x, y), \mathbf{n}(x, y) \rangle ds(x, y) \\ &= \int_{\overline{D_\varepsilon}} \nabla \cdot \mathbf{n}(x, y) dx dy + \int_{\partial B(\mathbf{0}, \varepsilon)} \langle \mathbf{n}(x, y), \frac{(x, y)}{\varepsilon} \rangle ds(x, y). \end{aligned}$$

若 $-\mathbf{n}(x, y)|_{\partial D}$ 在某一点从而处处指向 $\overline{D}$ 的外侧, 则

$$\begin{aligned} H_1(\partial D) &= \int_{\partial D} ds(x, y) = \int_{\partial D} \langle -\mathbf{n}(x, y), -\mathbf{n}(x, y) \rangle ds(x, y) \\ &= \int_{\overline{D_\varepsilon}} (-\nabla \cdot \mathbf{n}(x, y)) dx dy + \int_{\partial B(\mathbf{0}, \varepsilon)} \langle \mathbf{n}(x, y), \frac{(x, y)}{\varepsilon} \rangle ds(x, y). \end{aligned}$$

于是总有

$$H_1(\partial D) = \left| \int_{\overline{D_\varepsilon}} \nabla \cdot \mathbf{n}(x, y) dx dy \pm \int_{\partial B(\mathbf{0}, \varepsilon)} \langle \mathbf{n}(x, y), \frac{(x, y)}{\varepsilon} \rangle ds(x, y) \right|. \quad (*)$$

**计算  $\nabla \cdot \mathbf{n}(x, y)$ :** 注意梯度 $\nabla F(x, y)$ 是列向量, 由第八章第4节的习题中证明的关于计算单位向量映射微分的一个公式知

$$\begin{aligned} \mathbf{n}'(x, y) &= \left( \frac{\nabla F(x, y)}{|\nabla F(x, y)|} \right)' = \frac{1}{|\nabla F(x, y)|} \left( \mathbf{I} - \frac{\nabla F(x, y) \nabla F(x, y)^\tau}{|\nabla F(x, y)|^2} \right) (\nabla F)'(x, y) \\ &= \frac{1}{|\nabla F(x, y)|} \left( \mathbf{I} - \frac{\nabla F(x, y) \nabla F(x, y)^\tau}{|\nabla F(x, y)|^2} \right) \mathbf{H}_F(x, y), \quad (x, y) \in \mathbb{R}^2 \setminus \{\mathbf{0}\} \end{aligned}$$

其中

$$(\nabla F)'(x, y) = \mathbf{H}_F(x, y) \quad \text{是 } F \text{ 的 Hesse 矩阵.}$$

设 $\|\cdot\|$ 是矩阵的F-范数(即矩阵元素的平方和的开方), 则有

$$\begin{aligned} \|\mathbf{n}'(x, y)\| &\leq \left\| \frac{1}{|\nabla F(x, y)|} \mathbf{H}_F(x, y) \right\| + \left\| \frac{1}{|\nabla F(x, y)|^3} \nabla F(x, y) \nabla F(x, y)^\tau \mathbf{H}_F(x, y) \right\| \\ &\leq \frac{1}{|\nabla F(x, y)|} \|\mathbf{H}_F(x, y)\| + \frac{1}{|\nabla F(x, y)|^3} \|\nabla F(x, y)\| \|\nabla F(x, y)^\tau\| \|\mathbf{H}_F(x, y)\| \\ &= \frac{2}{|\nabla F(x, y)|} \|\mathbf{H}_F(x, y)\| \leq \frac{C}{(x^2 + y^2)^\alpha} \quad \forall (x, y) \in \overline{D}, \\ &\implies \\ |\nabla \cdot \mathbf{n}(x, y)| &= |\text{tr}(\mathbf{n}'(x, y))| \leq 2\|\mathbf{n}'(x, y)\| \leq \frac{2C}{(x^2 + y^2)^\alpha} \quad \forall (x, y) \in \overline{D}. \end{aligned}$$

这里 $0 < C < \infty$ 是常数, 并用到了 $\overline{D}$ 是紧集和 $F \in C^2(\mathbb{R}^2)$ . 易见

$$\lim_{\varepsilon \rightarrow 0^+} \nabla \cdot \mathbf{n}(x, y) \mathbf{1}_{\overline{D_\varepsilon}}(x, y) = \nabla \cdot \mathbf{n}(x, y) \quad \forall (x, y) \in \overline{D} \setminus \{\mathbf{0}\},$$

$$|\nabla \cdot \mathbf{n}(x, y) \mathbf{1}_{\overline{D_\varepsilon}}(x, y)| \leq \frac{2C}{(x^2 + y^2)^\alpha} \quad \forall (x, y) \in \overline{D}, \quad \forall 0 < \varepsilon < \varepsilon_0,$$

$$\int_{\overline{D}} \frac{2C}{(x^2 + y^2)^\alpha} dx dy \leq \int_{x^2 + y^2 \leq R^2} \frac{2C}{(x^2 + y^2)^\alpha} dx dy = 2\pi \int_0^R r \frac{2C}{r^{2\alpha}} dr = 4\pi C \frac{R^{2-2\alpha}}{2-2\alpha} < \infty.$$

因此据LDC (Lebesgue 控制收敛定理) 知

$$\int_{\overline{D}} |\nabla \cdot \mathbf{n}(x, y)| dx dy < \infty$$

且  $\lim_{\varepsilon \rightarrow 0^+} \int_{\overline{D_\varepsilon}} \nabla \cdot \mathbf{n}(x, y) dx dy = \int_{\overline{D}} \nabla \cdot \mathbf{n}(x, y) dx dy.$

又注意 $\mathbf{n}(x, y), \frac{(x, y)}{\varepsilon}$  都是单位向量有

$$\begin{aligned} \left| \int_{\partial B(\mathbf{0}, \varepsilon)} \left\langle \mathbf{n}(x, y), \frac{(x, y)}{\varepsilon} \right\rangle ds(x, y) \right| &\leq \int_{\partial B(\mathbf{0}, \varepsilon)} \left| \left\langle \mathbf{n}(x, y), \frac{(x, y)}{\varepsilon} \right\rangle \right| ds(x, y) \\ &\leq \int_{\partial B(\mathbf{0}, \varepsilon)} ds(x, y) = 2\pi\varepsilon \rightarrow 0 \quad (\varepsilon \rightarrow 0^+). \end{aligned}$$

所以在(\*)中令 $\varepsilon \rightarrow 0^+$  得到

$$H_1(\partial D) = \left| \int_{\overline{D}} \nabla \cdot \mathbf{n}(x, y) dx dy \right|.$$

□